Week 10: HPC Applications - Lab Guide

HPC & Cluster Computing

Objective: Run real-world scientific applications. **Goal**: Move beyond "Hello World" and run codes used by actual scientists.

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1. Prerequisites

- [ ] A working cluster with MPI.
- [ ] `openmpi` module loaded.

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2. Computational Fluid Dynamics (OpenFOAM)

We will run the standard "Cavity" tutorial: a lid-driven cavity flow simulation.

2.1 Install OpenFOAM

If OpenFOAM is not installed globally, we can pull a container (Warewulf/Singularity) or compile it (takes hours). For this lab, we assume a pre-compiled environment or container is available.

- [ ] Load the environment: `bash module load openfoam/v2212 source $FOAM_BASH`

2.2 Prepare the Case

Copy the tutorial to your home directory so you don't break the system files.

- [ ] Copy tutorial to `$FOAM_RUN`:

```
bash mkdir -p $FOAM_RUN cp -r $FOAM_TUTORIALS/incompressible/icoFoam/cavity/cavity $FOAM_RUN/ cd $FOAM_RUN/cavity
```

2.3 Mesh Generation

Create the mesh (the grid).

- [ ] Run `blockMesh`:

```
bash blockMesh
```

2.4 Run Simulation (Serial)

Run the solver on one core.

- [ ] Execute `icoFoam`:

```
bash icoFoam
```

2.5 Run Simulation (Parallel)

Now, let's go faster.

- [ ] **1. Edit Settings**: Open `system/decomposeParDict` and ensure `numberOfSubdomains` is set to `4`.
- [ ] **2. Decompose**: Chop the mesh into 4 pieces:

```
bash decomposePar
```

- [ ] **3. Run with MPI**:

```
bash mpirun -np 4 icoFoam -parallel
```

- [ ] **4. Reconstruct**: Stitch the results back together:

```
bash reconstructPar
```

- [ ] **Verification**: Check the `0.5`, `0.6`, etc. time directories. Do they exist?

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3. Bioinformatics (HMMER)

We will search a protein database for a specific sequence pattern.

3.1 Setup

- [ ] Load module and download data: `bash module load hmmer # Download a tiny database (tutorial data) wget ftp://ftp.ebi.ac.uk/pub/software/vertebrategenomics/hmmer/Tutorial.tar.gz tar xf Tutorial.tar.gz cd tutorial`

3.2 Run Search

Search for the "Globins" pattern in the database.

- [ ] Run `hmmsearch`:

```
bash hmmsearch globins4.hmm globins45.fa > results.txt
```

3.3 Verify

- [ ] Check the output: `bash head results.txt` *Did it find matches?*

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4. Submission

Submit a screenshot of:

- [ ] The `icoFoam` log showing the "ExecutionTime".
- [ ] The `results.txt` output from HMMER.

Course Index